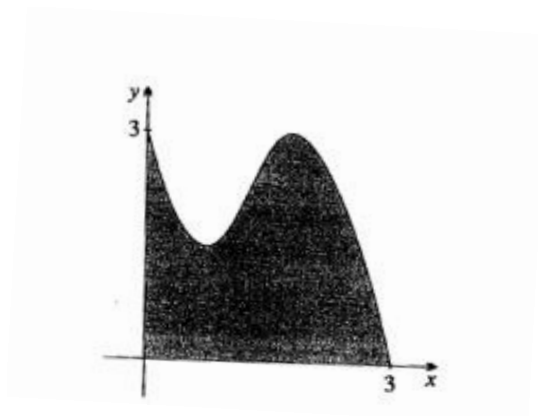


QUESTION 1 (12 marks)**Marks**

- a) 2



In the diagram above, the region under the curve $y = 3 - 4x + 4x^2 - x^3$ in the first quadrant is shaded. A solid of revolution is formed by rotating this region about the y -axis. Use the method of cylindrical shells to show the volume of this solid is given by

$$V = \pi \int_0^3 (6x - 8x^2 + 8x^3 - 2x^4) dx$$

- b) The region in the number plane that contains all points satisfying simultaneously the inequalities 4

$$x \leq 1, y \geq 1 \text{ and } y \leq e^x$$

is rotated through one complete revolution about the x -axis. Use the method of cylindrical shells to show that the volume of the resulting solid is

$$\frac{\pi}{2}(e^2 - 3).$$

- c) How many arrangements of six A's, five B's, and four C's are there in which the first A precedes the first B which precedes the first C? Explain your answer. 2

- d) A particle of mass m is moving vertically in a resisting medium in which the resistance to motion has magnitude $\frac{1}{10}mv^2$, when the particle has velocity $v \text{ ms}^{-1}$. The acceleration due to gravity is 10ms^{-2} .

- (i) The particle is projected vertically upwards with speed $U \text{ ms}^{-1}$. Show that during its upward motion, its acceleration $a \text{ ms}^{-2}$ is given by 1

$$a = -\frac{1}{10}(100 + v^2)$$

- (ii) Hence show that its maximum height, H metres, is given by 3

$$H = 5 \ln \left(\frac{U^2 + 100}{100} \right)$$

$P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ where $p, q > 0$ are two distinct points on the hyperbola H with equation $xy = c^2$. You are given that the equation of the tangent to H at $P\left(cp, \frac{c}{p}\right)$ is given by $x + p^2y = 2cp$ and that the normal to H at $P\left(cp, \frac{c}{p}\right)$ is given by $y = p^2x + \frac{c}{p} - cp^3$. DO NOT prove these results.

- (i) The tangents to H at $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ meet at T . Find the coordinates of T . 2
- (ii) Given that the chord PQ passes through the point $(2c, 0)$ find a relationship between the parameters p and q . 2
- (iii) Find the equation of the locus of T as P and Q move about H according to the restriction in part (ii). Give a complete description of this locus. 1
- (iv) The normal at P intersects the second 'branch' of the hyperbola at R . Show that the x coordinates of P and R are the roots of the equation 3

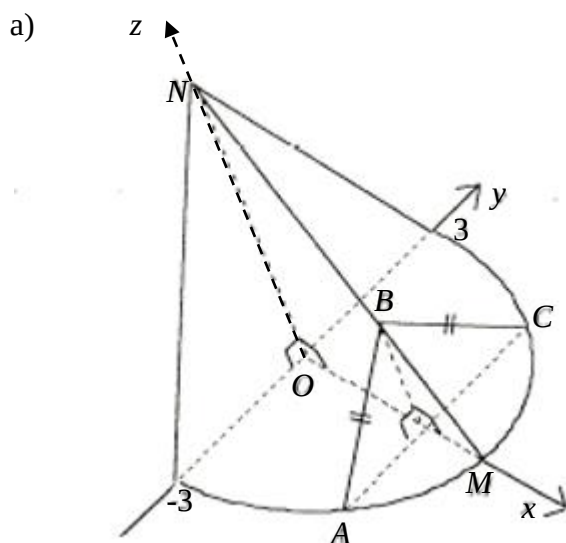
$$x^2 - c\left(p - \frac{1}{p^3}\right)x - \frac{c^2}{p^2} = 0$$

and find the coordinates of R .

- (v) The tangent at P intersects the x and y axes at A and B respectively. Show that the area of triangle ABR is given by 3

$$c^2\left(p^2 + \frac{1}{p^2}\right)^2$$

- (vi) Given the fact that the sum of any positive number and its reciprocal is greater than or equal to 2, find the minimum area of triangle ABR . 1



A solid figure has a semi-circular base of radius 3cm. Cross sections taken perpendicular to the x -axis are isosceles triangles. The vertical cross section containing the radius OM of the base of the solid is a right isosceles triangle OMN , where $OM = ON$.

- (i) Show that the area, A , of triangle ABC (where $AB = BC$) is given by 2

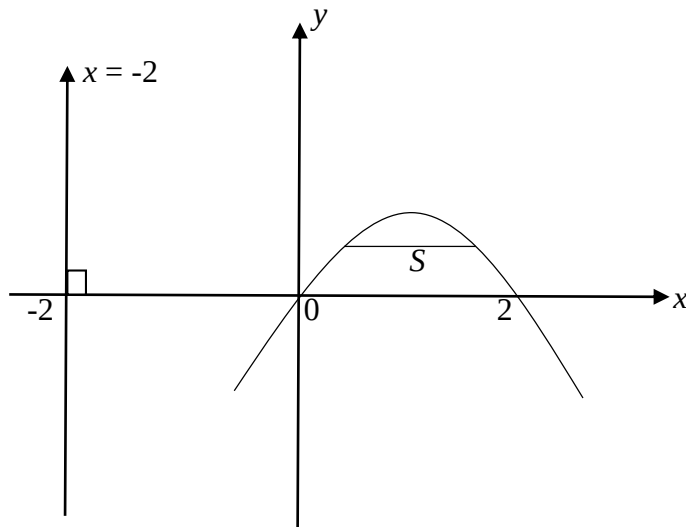
$$A = (3 - x)(9 - x^2)^{\frac{1}{2}}$$

- (ii) Find the volume of the solid. 4

- b) 21 books are to be distributed among 3 girls, Ava, Beth and Cate, so that Ava and Beth together receive twice as many books as Cate. Find the probability that each girl receives the same number of books. 3

- c) How many ways are there, on 6 successive nights, to enlist one of three siblings to help with dinner so that no one child is helping more than three times? Give a full explanation. 3

a)



The diagram shows the parabola $y = x(2 - x)$. When the region bounded by the parabola and the x -axis, $0 \leq x \leq 2$, is revolved about the line $x = -2$ to form a solid, the line segment S at height y sweeps out an annulus.

- (i) Show that the area of the annulus is given by $A = 12\pi\sqrt{1 - y}$. 3
 (ii) Hence find the volume of the solid formed. 2

- b) A particle of mass 2kg is propelled from the origin along the x -axis with an initial velocity of Q m/s. The only forces acting on the body in the direction of the x -axis are friction, which is a constant 16 Newtons, and air resistance which equals v^2 Newtons where v is the velocity of the particle t seconds after leaving the origin. 5

Show that

$$t = \frac{1}{2} \tan^{-1} \left(\frac{4Q - 4v}{16 + Qv} \right)$$

- c) A particle of mass m kg falls from rest in a medium where the resistance to motion is mkv when the particle has a velocity of v ms⁻¹ and k is a constant. Take gravity to be g ms⁻².

- (i) Find the terminal velocity, V , of the particle in this medium. 1
 (ii) Hence, show that the equation of motion of the particle is $\ddot{x} = k(V - v)$ where $\ddot{x}$ is acceleration. 1

A camera of mass m kg has a parachute attached. A cameraman releases the camera from rest at the top of a building 50 metres high. The forces acting during its fall are gravity and, owing to the parachute, a resistance force of $\frac{1}{10}mv^2$ when the speed of the camera is $v \text{ ms}^{-1}$.

Unfortunately, after $2\ln 2$ seconds, the parachute detaches itself from the camera, and then the only force acting on the camera is gravity. The acceleration due to gravity is taken as $g = 10 \text{ ms}^{-2}$. At time t seconds, the camera has fallen a distance x metres from the top of the building, and its speed is $v \text{ ms}^{-1}$.

- (i) Show that while the parachute is attached, $10\ddot{x} = 100 - v^2$ where $\ddot{x}$ is acceleration. 1

- (ii) Hence show that 6

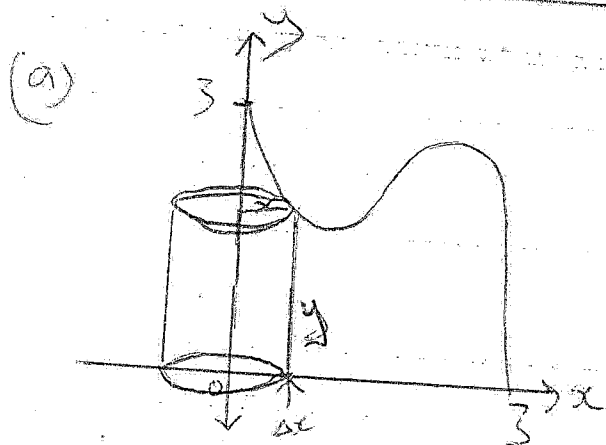
$$v = 10 \left(\frac{e^{2t} - 1}{e^{2t} + 1} \right)$$

and

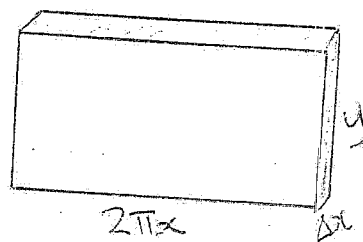
$$x = -5\ln \left[1 - \left(\frac{v}{10} \right)^2 \right]$$

- (iii) Find the exact speed of the camera and the exact distance fallen just before the parachute detaches. 2

- (iv) Find an expression for $\ddot{x}$ after the parachute detaches and find the speed of the camera just before it reaches the base of the building. Give your answer correct to 2 significant figures. 3



open the cylinder



$$\Delta V \doteq 2\pi x y \Delta x$$

but $y = 3 - 4x + 4x^2 - x^3$

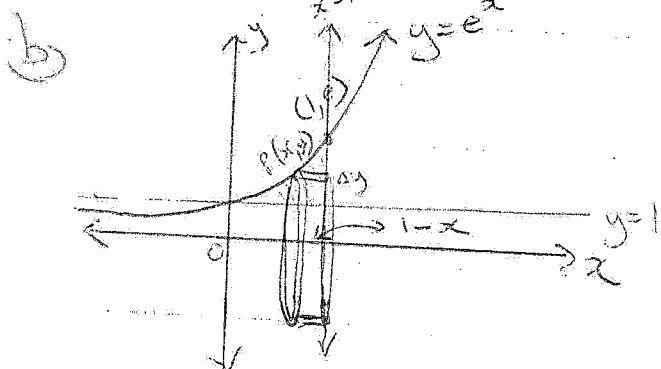
$$\begin{aligned} \therefore \Delta V &= 2\pi x (3 - 4x + 4x^2 - x^3) \Delta x \\ &= 2\pi (3x - 4x^2 + 4x^3 - x^4) \Delta x \end{aligned}$$

①

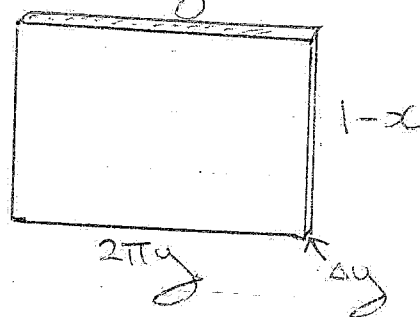
$$\begin{aligned} V &= \lim_{\Delta x \rightarrow 0} \sum_{x=0}^3 2\pi (3x - 4x^2 + 4x^3 - x^4) \Delta x \\ &= \pi \int_0^3 (6x - 8x^2 + 8x^3 - 2x^4) dx \end{aligned}$$

①

* Some diagrams were very disappointing, you needed to clearly show where your slice came from and its measurements, otherwise you lost the 1st mark.



unfold the cylinder



$$\Delta V \doteq 2\pi y (1-x) \Delta y$$

$$\Delta V = \lim_{\Delta y \rightarrow 0} \sum_{y=1}^e 2\pi y (1-x) \Delta y$$

$$\therefore V = 2\pi \int_1^e y(1-x) dy$$

①

$$= 2\pi \int_1^e y(1 - \ln y) dy = 2\pi \int_1^e y dy - 2\pi \int_1^e y \ln y dy$$

2/5

let $u = \ln y$ $v' = \frac{1}{y}$
 $u' = \frac{1}{y}$ $v = \frac{1}{2} y^2$

$$\begin{aligned}
 V &= 2\pi \left[\frac{y^2}{2} \right]_1^e - 2\pi \left\{ \left[\frac{1}{2} y^2 \ln y \right]_1^e - \int_1^e \frac{1}{2} y dy \right\} \\
 &= 2\pi \left(\frac{e^2}{2} - \frac{1}{2} \right) - 2\pi \left\{ \left(\frac{1}{2} e^2 - 0 \right) - \frac{1}{2} \left[\frac{y^2}{2} \right]_1^e \right\} \\
 &= 2\pi \left(\frac{e^2}{2} - \frac{1}{2} \right) - 2\pi \left(\frac{1}{2} e^2 - \frac{1}{2} \left(\frac{e^2}{2} - \frac{1}{2} \right) \right) \\
 &= 2\pi \left(\frac{e^2}{2} - \frac{1}{2} - \frac{e^2}{2} + \frac{e^2}{4} - \frac{1}{4} \right) \\
 &= 2\pi \left(\frac{e^2}{4} - \frac{3}{4} \right) \\
 &= \frac{\pi}{2} (e^2 - 3)
 \end{aligned}$$

2nd mark for correctly doing the Integration by parts
 3rd mark for putting it altogether and factoring 2π .
 4th mark for simplifying correctly without fudging.
 * If you did it by slices then it was a maximum of 2 marks.

(c) no. of arrangements = $1 \times {}^{14}C_5 \times 1 \times \frac{8!}{4!4!}$

$\swarrow$ must start with an A
 $\downarrow$ place the other A's
 $\swarrow$ B must go first in the remaining places.
 $\rightarrow$ place the rest of B's and C's

$= 1 \times$
 $= 140140$

alternative ①

$\frac{3}{5}$

$$\begin{array}{l} A, B, \\ \quad \quad \quad \frac{13!}{4!5!4!} \quad 90090 \\ \quad \quad \quad (C's) (A's) (B's) \\ A, AB, \\ \quad \quad \quad + \frac{12!}{4!4!4!} \quad + 34650 \\ A, AAB, \\ \quad \quad \quad + \frac{11!}{4!3!4!} \quad + 11550 \\ A, AAAB, \\ \quad \quad \quad + \frac{10!}{4!2!4!} \quad + 3150 \\ A, AAAAB, \\ \quad \quad \quad + \frac{9!}{4!1!4!} \quad + 630 \\ A, AAAAAB, \\ \quad \quad \quad + \frac{8!}{4!4!} \quad + 70 \\ \downarrow \\ = 140140 \end{array}$$

Since only matters when the first B occurs.

Alternative ②

A ① A ② A ③ A ④ A ⑤ A ⑥

ie 6 Gaps

case 1 : 1st B in 1st gap

- ways to replace remaining B's = $\frac{6 \times 7 \times 8 \times 9}{4!} = 126$

- ways to place the 4 C's = $\frac{10 \times 11 \times 12 \times 13}{4!} = 715$

Total for case 1 = $126 \times 715 = 90090$ ways

case 2 : 1st B in second gap

- ways to replace remaining B's = $\frac{5 \times 6 \times 7 \times 8}{4!} = 70$

- ways to place the 4 C's = $\frac{9 \times 10 \times 11 \times 12}{4!} = 495$

Total for case 2 = $70 \times 495 = 34650$

case 3 : 1st B in 3rd gap

- ways to replace remaining B's = $\frac{4 \times 5 \times 6 \times 7}{4!} = 35$

- ways to place the 4 C's = $\frac{8 \times 9 \times 10 \times 11}{4!} = 330$

Total for case 3 = $35 \times 330 = 11550$ ways

4/5

Case 4: 1st B in 4th gap

- ways to place the remaining B's = $\frac{3 \times 4 \times 5 \times 6}{4!} = 15$
- ways to place the 4 C's = $\frac{7 \times 8 \times 9 \times 10}{4!} = 210$

Total for case 4 = $15 \times 210 = 3150$

Case 5: 1st B in 5th gap

- ways to place the remaining B's = $\frac{2 \times 3 \times 4 \times 5}{4!} = 5$
- ways to place the 4 C's = $\frac{6 \times 7 \times 8 \times 9}{4!} = 126$

Total for case 5 = $5 \times 126 = 630$ ways

Case 6: 1st B in 6th gap

- ways to place the remaining B's = 1
- ways to place the 4 C's = $\frac{5 \times 6 \times 7 \times 8}{4!} = 70$

Total = 70

Total number of ways = $90090 + 34650 + 11550 + 3150 + 630 + 70 = 140140$

* If you attempted alternative ① or ② and showed working with an explanation, then you scored 1 mark if you had a correct case.

+ If you misunderstood the question, and therefore made it easier, your answer was 27720, then you scored a maximum of one mark, as long as you had an explanation.

in 27720 with no words = 0 marks

* 105105 = 0 marks.

Alternate ③

sit A, B and C

- There are 14 places that A can choose,
- $${}^{14}C_5 \times {}^8C_4 \times {}^4C_4 = 140140$$

d) (i)

$$\begin{aligned} t=0 \\ x=0 \\ v=u \end{aligned}$$

Forces on particle
 $\frac{1}{10}mv^2$
 $10m$

if "m" is missing they lose one mark!

$$m\ddot{x} = -(10m + \frac{1}{10}mv^2)$$

$$= -\frac{1}{10}(100m + mv^2)$$

①

$$\therefore \ddot{x} = -\frac{1}{10}(100 + v^2)$$

$$\text{i.e. } a = -\frac{1}{10}(100 + v^2)$$

$$(ii) \quad v \frac{dv}{dx} = -\frac{1}{10}(100 + v^2)$$

$$\text{or } v \frac{dv}{dx} = -\frac{1}{10}(100 + v^2)$$

$$\frac{dx}{dv} = \frac{-10v}{100 + v^2}$$

$$\int_u^H \frac{v dv}{100 + v^2} = -\frac{1}{10} \int_0^H dx \quad \text{①}$$

$$= -5 \left(\frac{2v}{100 + v^2} \right)$$

$$\left[\frac{1}{2} \ln(100 + v^2) \right]_u^H = -\frac{1}{10} [x]_0^H$$

①

$$x = -5 \ln(100 + v^2) + C \quad \text{①}$$

$$x=0, v=u$$

$$\frac{1}{2} [\ln(100) - \ln(100 + u^2)] = -\frac{1}{10} H$$

$$0 = -5 \ln(100 + u^2) + C \quad -\frac{10}{2} (\ln 100 - \ln(100 + u^2)) = H$$

$$C = +5 \ln(100 + u^2)$$

$$\therefore H = 5 (\ln(100 + u^2) - \ln 100)$$

①

$$\therefore x = -5 \ln(100 + v^2) + 5 \ln(100 + u^2)$$

$$= 5 \ln \left(\frac{100 + u^2}{100 + v^2} \right)$$

$$\therefore H = 5 \ln \left(\frac{u^2 + 100}{100} \right)$$

At max height, H, v=0

①

$$\therefore H = 5 \ln \left(\frac{u^2 + 100}{100 + 0} \right)$$

$$= 5 \ln \left(\frac{u^2 + 100}{100} \right)$$

for this method students had to show the substitution

Suggested Solutions

Marks

Marker's Comments

(i) Tangent at P : $x + p^2 y = 2cp \dots (1)$

" " Q : $x + q^2 y = 2cq \dots (2)$

$(1) - (2) : y(p^2 - q^2) = 2c(p - q)$

$$y = \frac{2c(p - q)}{(p - q)(p + q)}$$

$$= \frac{2c}{p + q} \dots (3)$$

Sub (3) into (1)

$$x = -p^2 \left(\frac{2c}{p + q} \right) + 2cp$$

$$= \frac{-2cp^2 + 2cp^2 + 2cpq}{p + q}$$

$$= \frac{2cpq}{p + q}$$

$$\therefore T \left(\frac{2cpq}{p + q}, \frac{2c}{p + q} \right)$$

1

For correct x-coordinate

1

For correct y-coordinate

(ii) Equation of PQ : $y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$

$$\Rightarrow y - \frac{c}{p} = -\frac{1}{pq} (x - cp)$$

Sub $(2c, 0)$:

$$-\frac{c}{p} = -\frac{1}{pq} (2c - cp)$$

$$\Rightarrow p + q = 2$$

1

For equation of PQ

1

For correct relationship

Suggested Solutions

Marks

Marker's Comments

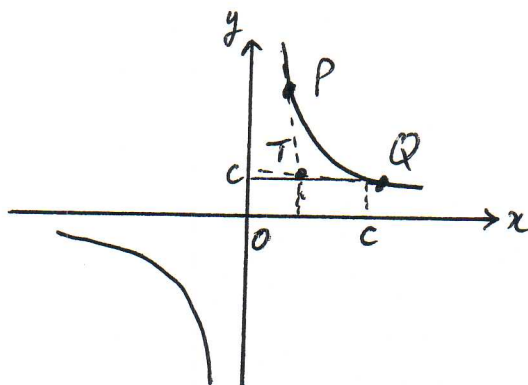
(iii) From (i) $y = \frac{2c}{p+q}$ and from (ii),

$$p+q = 2$$

$$\therefore y = \frac{2c}{2}$$

$$= c$$

But $p, q > 0$ and T cannot possibly be on $xy = c^2$ as p and q are distinct. Hence the locus is $y = c$ with $0 < x < c$



(iv) (a): Normal at P: $y = p^2x + \frac{c}{p} - cp^3$

Sub $y = c^2/x$ into above

$$\Rightarrow c^2 = p^2x^2 + \frac{c}{p}x - cp^3x$$

$$p: \quad \frac{c^2}{p^2} = x^2 + \frac{cx}{p^3} - cp^2x$$

$$\therefore x^2 - c(p - \frac{1}{p^3})x - \frac{c^2}{p^2} = 0$$

1 For both the locus equation and its restriction

1 For substitution and simplification and obtaining the desired result.

Suggested Solutions

Marks

Marker's Comments

(iv) (β): continued:

$$\text{from } x^2 - c(p - \frac{1}{p^3})x - \frac{c^2}{p^2} = 0$$

$$\begin{aligned} \text{Sum of roots, } x_1 + x_2 &= -\frac{b}{a} \\ &= c(p - \frac{1}{p^3}) \end{aligned}$$

$$\text{But } x_1 = cp = p_x$$

$$\begin{aligned} \therefore x_2 &= cp - \frac{c}{p^3} - cp \\ &= -\frac{c}{p^3} = R_x \end{aligned}$$

$$\text{Sub } x = -\frac{c}{p^3} \text{ into } y = \frac{c^2}{x^2}$$

$$\therefore y = -cp^3$$

$$\therefore R(-\frac{c}{p^3}, -cp^3)$$

$$\underline{\text{OR}}: x^2 - c(p - \frac{1}{p^3})x - \frac{c^2}{p^2} = 0$$

$$\Rightarrow (x - cp)(x + \frac{c}{p^3}) = 0$$

$$\underline{\text{OR}}: x = \frac{c(p - \frac{1}{p^3}) \pm \sqrt{c^2(p - \frac{1}{p^3})^2 + 4\frac{c^2}{p^2}}}{2}$$

1+1 For each coordinate of R

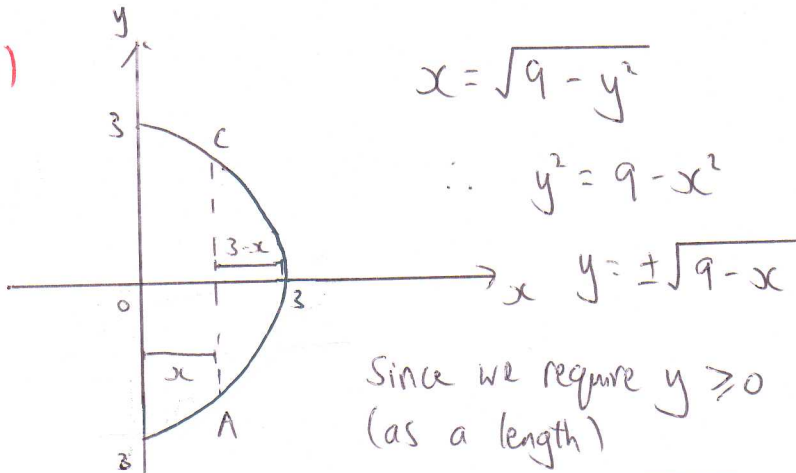
Some alternate ways of R_x

Suggested Solutions

Marks

Marker's Comments

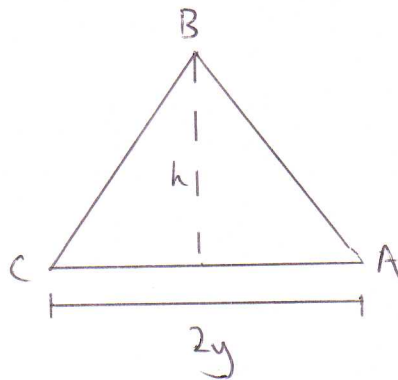
a) i)



$$y = \sqrt{9 - x^2}$$

①

Some explanation



Since $h = 3 - x$
(vertical cross section is
an Isosceles triangle)

①

Some explanation/
diagram

$$\text{then } A = \frac{1}{2} \times h \times 2y$$

$$= \frac{1}{2} (3 - x) \times 2\sqrt{9 - x^2}$$

$$A = (3 - x)\sqrt{9 - x^2}$$

or

$h = 3 - x$ could be proven by

— Similar Triangles

— finding the linear equation from M to N
getting $z = 3 - x$.

Suggested Solutions	Marks	Marker's Comments
Q ii) $V = \lim_{\Delta x \rightarrow 0} \sum_0^3 (3-x)(9-x^2)^{\frac{1}{2}} \Delta x$ $= \int_0^3 (3-x)(9-x^2)^{\frac{1}{2}} dx$ ————— ① $= \int_0^3 3(9-x^2)^{\frac{1}{2}} dx - \int_0^3 x(9-x^2)^{\frac{1}{2}} dx$ $= 3 \times \text{Area (quadrant radius 3)} + \frac{1}{2} \int_0^3 (-2x)(9-x^2)^{\frac{1}{2}} dx$ ① ————— $= \frac{27\pi}{4} + \frac{1}{2} \left[\frac{(9-x^2)^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^3$ ————— ① value $= \frac{27\pi}{4} + \left(0 - \frac{9(3)}{3} \right)$ $= \frac{27\pi}{4} - 9$ ————— ①		Integration
Students who used Trig Substitution: ① Writing the integral ① Writing the integral in terms of θ and no square roots i.e. $\int_0^{\frac{\pi}{2}} (3-3\sin\theta) \times 9\cos^2\theta d\theta$ ① Integrate ① Evaluate		

Suggested Solutions

Marks

Marker's Comments

b) Ava and Beth receive 14
Cate receives 7

①

Remaining 2 marks :

Method 1

Number of ways to distribute into (7,7,7)

$$= {}^{21}C_{14} \times {}^{14}C_7 = 399072960$$

①

Expression will suffice

Ways to choose 14 for Ava and Beth
Ways to choose 7 to belong to Ava

Number of ways to distribute into (7,14)

$$= {}^{21}C_{14} \times (2^{14} - 2)$$

Ways to choose 14 for Ava and Beth
ways to divide 14 into 2 group each person gets something!!!

$$\therefore P = \frac{{}^{21}C_{14} \times {}^{14}C_7}{{}^{21}C_{14} \times (2^{14} - 2)} = \frac{{}^{14}C_7}{2^{14} - 2} = \frac{1716}{8190}$$

or

$$0.209498$$

①

If you forget to take away 2 from denominator you will get $\frac{429}{2048}$ or 0.209472

Wrong!!!

Suggested Solutions

Marks

Marker's Comments

Method 2

$$P(7,7,7) = \frac{21!}{7!7!7!} \left(\frac{1}{3}\right)^7 \left(\frac{1}{3}\right)^7 \left(\frac{1}{3}\right)^7$$

$$P(7,14) = \frac{21!}{7!14!} \left(\frac{1}{3}\right)^7 \left(\frac{2}{3}\right)^{14}$$

$$\begin{aligned} \therefore P &= \frac{\frac{21!}{7!7!7!} \left(\frac{1}{3}\right)^{21}}{\frac{21!}{7!14!} \left(\frac{1}{3}\right)^7 \left(\frac{2}{3}\right)^{14}} \\ &= \frac{14!}{7!7!} \times \frac{\left(\frac{1}{3}\right)^{14}}{\left(\frac{2}{3}\right)^{14}} \\ &= \frac{{}^{14}C_7 \left(\frac{1}{3}\right)^{14}}{2^{14} \left(\frac{1}{3}\right)^{14}} \\ &= \frac{{}^{14}C_7}{2^{14}} \end{aligned}$$

Since sample space (denominator) must subtract 2 as explained above.

$$P = \frac{{}^{14}C_7}{2^{14} - 2} = \frac{1716}{8191}$$

①

①

Suggested Solutions

Marks

Marker's Comments

c) Method 1

- Each child helps out 2 nights (2, 2, 2)

$$= \frac{6!}{2!2!2!} \text{ — Ways to line 6 up in order}$$

$$= \frac{6!}{2!2!2!} \text{ — 3 repeats of 2}$$

$$= \underline{\underline{90}}$$

- One child one night, one child two nights, one child 3 nights (1, 2, 3)

$$\text{Ways to line 6 up} = \frac{6!}{3!2!} \times 3! \text{ — Ways to arrange who does 1, 2, 3 nights.}$$

$$\text{a repeat of 3 and a repeat of 2} = \underline{\underline{360}}$$

- 2 Children help 3 nights each

$$\text{line 6 up} = \frac{6!}{3!3!} \times 3 \text{ — Ways to choose who misses out.}$$

$$\text{2 repeats of 3.} = \underline{\underline{60}}$$

$$\therefore \text{Total} = 90 + 360 + 60$$

$$= \underline{\underline{510}}$$

Suggested Solutions

Marks

Marker's Comments

c) Method 2

$$\begin{aligned} \text{Total number of arrangements with no restrictions} \\ &= 3^6 \\ &= \underline{\underline{729}} \end{aligned}$$

$$\text{Number (1 person does 6 nights)} = \underline{\underline{3}} \quad \text{--- 3 different children}$$

$$\begin{aligned} \text{Number (1 does 5 nights)} &= 3 \times {}^6C_5 \times 2 \quad \text{--- 2 children to do the last night.} \\ &\quad \text{3 different child to do 5 nights} \quad \text{choose 5 nights from 6} \\ &= \underline{\underline{36}} \end{aligned}$$

$$\begin{aligned} \text{Number (1 does 4 nights)} &= 3 \times {}^6C_4 \times \underline{2 \times 2} \\ &\quad \text{3 different child to do 4 nights} \quad \text{choose 4 nights from 6} \quad \text{2 children to choose to do 2 nights} \\ &= \underline{\underline{180}} \end{aligned}$$

$$\begin{aligned} \therefore \text{Answer} &= 729 - 3 - 36 - 180 \\ &= \underline{\underline{510}} \end{aligned}$$

1st mark - Getting 729 AND one of 3, 36 or 180

2nd mark - Getting a 2nd restriction

3rd mark - Final answer.

MATHEMATICS Extension 2: Question...4...

Suggested Solutions

Marks

Marker's Comments

4. a) i) $y = 2x - x^2$

$-y = x^2 - 2x$

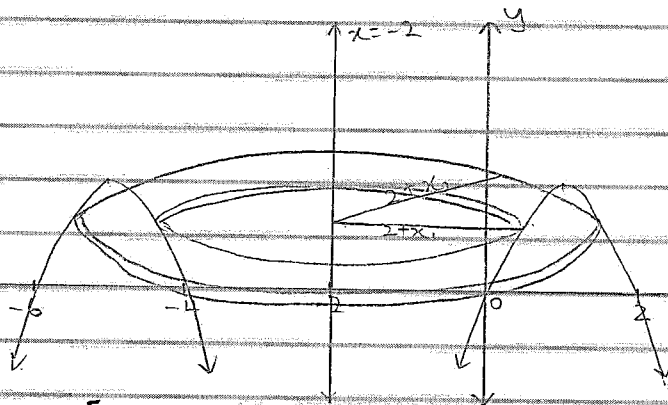
$1 - y = x^2 - 2x + 1$

$1 - y = (x - 1)^2$

$x - 1 = \pm \sqrt{1 - y}$

$x = 1 \pm \sqrt{1 - y}$

$x_2 = 1 + \sqrt{1 - y}$
 $x_1 = 1 - \sqrt{1 - y}$



$$\begin{aligned} A &= \pi [(2 + x_2)^2 - (2 + x_1)^2] \\ &= \pi [(2 + 1 + \sqrt{1 - y})^2 - (2 + 1 - \sqrt{1 - y})^2] \\ &= \pi [(3 + \sqrt{1 - y})^2 - (3 - \sqrt{1 - y})^2] \\ &= \pi [(3 + \sqrt{1 - y} + 3 - \sqrt{1 - y})(3 + \sqrt{1 - y} - (3 - \sqrt{1 - y}))] \\ &= \pi [(6)(2\sqrt{1 - y})] \\ &= 12\pi \sqrt{1 - y} \end{aligned}$$

OR

$1 - x = \sqrt{1 - y}$

$$\begin{aligned} A &= \pi [(4 - x)^2 - (2 + x)^2] \\ &= \pi [16 - 8x + x^2 - 4 - 4x - x^2] \\ &= \pi (12 - 12x) \\ &= 12\pi (1 - x) \\ &= 12\pi \sqrt{1 - y} \end{aligned}$$

ii) $V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^1 12\pi \sqrt{1 - y} \Delta y$
 $= 12\pi \int_0^1 \sqrt{1 - y} dy$
 $= 12\pi \left[-\frac{2}{3} \times (1 - y)^{3/2} \right]_0^1$
 $= -8\pi (0 - 1) = 8\pi$

show correct equation and substitution

correct expansion and work

correct equation and expansion

correct working and substitution.

① limits

① correct integration and solution.

MATHEMATICS Extension 2: Question...4...

Suggested Solutions

Marks

Marker's Comments

b) $F = ma$

$$\therefore 2\ddot{x} = -16 - v^2$$

$$\ddot{x} = \frac{-16 - v^2}{2}$$

$$\frac{2\ddot{x}}{16 + v^2}$$

$$\frac{dv}{dt} = -\frac{1}{2}(16 + v^2)$$

$$\int_0^v \frac{dv}{16 + v^2} = -\frac{1}{2} \int_0^t dt$$

$$\frac{1}{4} [\tan^{-1} \frac{v}{4}]_0^v = -\frac{1}{2} [t]_0^t$$

$$t = -\frac{1}{2} [\tan^{-1} \frac{v}{4} - \tan^{-1} \frac{0}{4}]$$

$$t = \frac{1}{2} [\tan^{-1} \frac{0}{4} - \tan^{-1} \frac{v}{4}]$$

$$2t = \tan^{-1}(\frac{0}{4}) - \tan^{-1}(\frac{v}{4})$$

$$\tan(2t) = \tan[\tan^{-1}(\frac{0}{4}) - \tan^{-1}(\frac{v}{4})]$$

$$= \frac{\tan(\tan^{-1} \frac{0}{4}) - \tan(\tan^{-1} \frac{v}{4})}{1 + \tan(\tan^{-1} \frac{0}{4}) \tan(\tan^{-1} \frac{v}{4})}$$

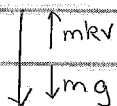
$$= \frac{\frac{0}{4} - \frac{v}{4}}{1 + \frac{0}{4} \times \frac{v}{4}} \times \frac{16}{16}$$

$$= \frac{4Q - 4v}{16 + Qv}$$

$$\therefore 2t = \tan^{-1} \left(\frac{4Q - 4v}{16 + Qv} \right)$$

$$t = \frac{1}{2} \tan^{-1} \left(\frac{4Q - 4v}{16 + Qv} \right)$$

c) i)



$$m\ddot{x} = mg - mkv$$

$$\ddot{x} = g - kv$$

for terminal velocity V , $\ddot{x} = 0$

$$\therefore 0 = g - kv$$

$$\therefore v = \frac{g}{k}$$

① correct limits.

① correct integration and formula for t .

① taking $\tan$ of both sides

① getting $\frac{\frac{0}{4} - \frac{v}{4}}{1 + \frac{0}{4} \times \frac{v}{4}}$ and getting final solution.

MATHEMATICS Extension 2: Question...4..

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} \text{ii) } \ddot{x} &= g - kv \\ \text{from (i)} \\ v &= \frac{g}{k} \\ \therefore g &= kv \\ \therefore \ddot{x} &= kv - kv \\ &= k(v - v) \end{aligned}$$

1

Suggested Solutions

Marks

Marker's Comments

(i)

$$m\ddot{x} = mg - \frac{1}{10}mv^2$$

$$\ddot{x} = 10 - \frac{1}{10}v^2$$

$$\underline{10\ddot{x} = 100 - v^2}$$

(ii) $10 \frac{dv}{dt} = 100 - v^2$

$$\frac{dt}{dv} = \frac{10}{100 - v^2}$$

Now $\frac{10}{100 - v^2} = \frac{a}{10 - v} + \frac{b}{10 + v}$

$$10 = a(10 + v) + b(10 - v)$$

$V=10 \quad 20a = 10$

$$a = \frac{1}{2}$$

$V=0 \quad 10 = 10a + 10b$

$$10 = 5 + 10b$$

$$\therefore b = \frac{1}{2}$$

$$\therefore \int_0^t dt = \frac{1}{2} \int_0^v \frac{1}{10+v} + \frac{1}{10-v} dv$$

Had to have
a diagram
clearly indicating
positive direction

cannot
quote this.

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} \text{OR} \quad \frac{10}{100-v^2} &= \frac{1}{2} \left(\frac{10+v+10-v}{(10-v)(10+v)} \right) \\ &= \frac{1}{2} \left[\frac{1}{10-v} + \frac{1}{10+v} \right] \end{aligned}$$

1

If no indication of where $\frac{1}{2}$ curve from left this, mark

$$\therefore t = \frac{1}{2} \left[\ln(10+v) - \ln(10-v) \right]$$

$$= \frac{1}{2} (\ln 10 - \ln 10)$$

$$= \frac{1}{2} \ln \left(\frac{10+v}{10-v} \right)$$

$$2t = \ln \left(\frac{10+v}{10-v} \right)$$

1

$$e^{2t} = \frac{10+v}{10-v}$$

$$10e^{2t} - ve^{2t} = 10 + v$$

$$v(e^{2t} + 1) = 10(e^{2t} - 1)$$

$$v = 10 \left(\frac{e^{2t} - 1}{e^{2t} + 1} \right)$$

1

Suggested Solutions

Marks

Marker's Comments

(ii)

$$10 v \frac{dv}{dx} = 100 - v^2$$

$$\frac{dv}{dx} = \frac{100 - v^2}{10v}$$

$$\frac{dx}{dv} = \frac{10}{100 - v^2}$$

$$\int_0^x \frac{10v}{100 - v^2} dv = \int_0^x dx$$

$$\therefore \left[-5 \ln(100 - v^2) \right]_0^x = x$$

$$-5 \ln(100 - v^2) - 5 \ln 100 = x$$

$$x = -5 \ln \left(\frac{100 - v^2}{100} \right)$$

$$= -5 \ln \left(1 - \frac{v^2}{100} \right)$$

 (iii) when $t = 2 \ln 2$

$$v = 10 \left(\frac{e^{4 \ln 2} - 1}{e^{4 \ln 2} + 1} \right)$$

$$= 10 \left(\frac{16 - 1}{16 + 1} \right)$$

$$= 150/17 \text{ m/s}$$

1

1

1

1

Should have put this step in but not penalised

Student is calculating $e^{2 \ln 2}$ for $t = 2$

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned}
 x &= -5 \ln \left(1 - \left(\frac{150}{17} \right)^2 \right) \\
 &= -5 \ln \frac{64}{289} \\
 &\text{or } 10 \ln \frac{17}{8} \text{ m.}
 \end{aligned}$$

(iv)

$$\begin{aligned}
 \downarrow x \quad \downarrow mg \quad m\ddot{x} &= mg \\
 \ddot{x} &= 10
 \end{aligned}$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 10$$

$$\frac{1}{2} v^2 = 10x + C_3$$

$$\text{When } x = 10 \ln \frac{17}{8} \quad v = \frac{150}{17}$$

$$C_3 = -36.4498$$

$$v^2 = 20x - 72.8996$$

$$\text{OR } \left\{ \begin{array}{l} \text{When } x = 50 \\ v^2 = 20 \times 50 - 72.8996 \\ v = 30.11 \end{array} \right.$$

$$v = 30.11$$

$$v = 30.11$$

$$\text{OR } \int_{10 \ln \frac{17}{8}}^{50} x dx = \int_{\frac{150}{17}}^v v dv$$

$$\frac{v^2}{2} - \left(\frac{150}{17} \right)^2 = 10 \left(50 + 10 \ln \frac{17}{8} \right)$$

$$= 30.11$$

Students had
 1-correct
 limits
 those who
 used to
 instead of
 x were
 unsuccessful